Sushmitha K G

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Summary

Computer Science engineering student with experience in ML model development, data analysis and AI-driven automation, skilled in Python, C and modern web technologies. Passionate about advanced algorithms, system integration and translating research concepts into practical, real-world solutions.

Education

P A College Of Engineering Mangalore — 2022 – 2026

CGPA-9

Bachelor of Engineering in Computer Science And Engineering

• Relevant Coursework: Data Structures and Algorithms, Operating Systems, Com.puter Networks, DBMS, Machine Learning, Artificial Intelligence, Blockchain and computer vision

Projects

AI-Based Brain Tumor Detection & Classification

- Designed a deep learning model using Swin Transformer for accurate tumor detection from MRI scans.
- Implemented preprocessing, feature extraction, and classification workflows.
- Added Grad-CAM explainability to highlight tumor regions and improve clinical trust.

Blockchain-Based Voting System

- Developed a secure online voting platform using Ethereum and smart contracts.
- Ensured transparency, immutability, and voter anonymity through blockchain architecture.
- Designed a reliable user authentication flow to support safe and seamless voting.

Automatic Waste Segregation System

- Built an embedded system using Arduino and sensors to detect wet and dry waste.
- Automated waste classification to improve speed and accuracy in segregation.
- Selected as a finalist at the ACCS Design Challenge 2024 for innovation and effectiveness.

Technical Skills

Languages / Frameworks: Python, JavaScript, React, HTML, CSS

Technologies / Tools: SQL, GitHub, VS Code, Arduino IDE

Certifications and Achievements

- Completed CS50x (Computer Science) from Harvard University (edX).
- Completed Big Data 101 and Hadoop 101 certifications from IBM Cognitive Class.
- Certified in Python, C Programming, and Data Visualization via Infosys Springboard.
- Completed Networking Basics certification from Cisco Networking Academy.
- Earned Data Structures in C c,e rtificate from Great Learning Academy. Participated in Flutter Forward Workshop by Google Developer Student Clubs (GDSC).
- Participated in GirlGeekHack'23 Hackathon, organized by IEEE Computer Society & NITK Surathkal.
- Participated in ICRAI 2024 – International Conference on Responsible AI, Mangalore University.
- Finalist in the ACCS Design Challenge 2024 at IIIT Bangalore for Smart Waste Segregation System. Participated in Hack Summit 2025, developing an AI-Driven Scholarship System (24-hour hackathon).

Paper Presented

"Optimized Deep Learning Model for Multi-Class Student Performance Analysis"

Co-Author & Presenter, ERCICAM 2025 – 2nd International Conference

Presented a research paper focusing on advanced machine learning techniques for analyzing and predicting student performance using optimized deep learning architectures.